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GitHub Link	https://github.com/mrclassic7499/MLA0405-DEEP-LEARNING/tree/main/Assessment/CO5

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 5: Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

Course Outcome Weightage: 35%
Assessment Tool Weightage: 20%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I Shaik Mubarak(192425194) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: Shaik Mubarak

Criteria	Subject Knowledge and Conceptual Understanding (3)	Technical Accuracy and Application of Concepts (2.5)	Problem-Solving and Analytical Thinking (2)	Communication Skills and Clarity of Explanation (1.5)	Confidence, Presentation and Professional Behaviour (1)	Total (10)
Weightage	30 %	25%	20%	15 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						

Q8						
Q9						
Q10						
Q11						
Q12						
Q13						
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Q15						
Q16						
Q17						
Q18						
Q19						
Q20						
Q21						
Q22						
Q23						
Q24						
Q25						

Signature of the Faculty:

Total Marks:

QUESTIONS: 25

Total Marks:25

1. What is an Autoencoder in Deep Learning?

Answer: An Autoencoder is a neural network that learns to compress input data and reconstruct it back.

2. What is the main purpose of an Autoencoder?

Answer: Its main purpose is to learn important features from data and reproduce the original input.

3. What are the major components of an Autoencoder?

Answer: The major components are:

- Encoder
- Bottleneck layer
- Decoder

4. What is the role of the Encoder in an Autoencoder?

Answer: The Encoder converts the input data into a smaller, compressed representation.

5. What is the role of the Decoder in an Autoencoder?

Answer: The Decoder reconstructs the original input from the compressed representation.

6. What is meant by the Bottleneck layer in an Autoencoder?

Answer: It is the middle layer that stores the most important features of the input in compressed form.

7. What is meant by Reconstruction in an Autoencoder?

Answer: Reconstruction is the process of reproducing an output that is as close as possible to the original input.

8. How does an Autoencoder learn to reconstruct its input?

Answer: It compares the reconstructed output with the original input and minimizes the reconstruction error during training.

9. What is the difference between an Autoencoder and a traditional Neural Network?

Answer: An Autoencoder learns to reproduce its input, while a traditional neural network usually predicts a specific output or class.

10. What type of loss function is commonly used to train an Autoencoder?

Answer: Mean Squared Error (MSE) is commonly used to measure the difference between the input and reconstructed output.

11. Explain the basic architecture of an Autoencoder with a suitable example.

Answer: The basic architecture is:

Input → Encoder → Bottleneck → Decoder → Reconstructed Output

Example: 784 Input Neurons → 128 → 32 → 128 → 784 Output Neurons.

12. Why is the dimensionality of the bottleneck layer usually smaller than the input layer?

Answer: It forces the network to learn only the important features and remove unnecessary information.

13. What happens if the bottleneck layer has too many neurons?

Answer: The model may simply copy the input and may not learn meaningful features.

14. What happens if the bottleneck layer has very few neurons?

Answer: Too much information may be lost, resulting in poor reconstruction.

15. What is a Denoising Autoencoder?

Answer: A Denoising Autoencoder is trained to reconstruct clean data from noisy or corrupted input.

16. How can an Autoencoder be used for image denoising?

Answer: A noisy image is given as input, and the Autoencoder learns to produce a clean version of the image.

17. How can Autoencoders be used for dimensionality reduction?

Answer: The Encoder compresses high-dimensional data into a smaller latent representation while retaining important features.

18. How can an Autoencoder be used for anomaly detection?

Answer: It reconstructs normal data accurately, but abnormal data produces a high reconstruction error, helping detect anomalies.

19. What is the difference between a Denoising Autoencoder and a basic Autoencoder?

Answer: A basic Autoencoder reconstructs the original input, while a Denoising Autoencoder reconstructs clean data from noisy input.

20. Give some real-world applications of Autoencoders.

Answer: Applications include:

- Image denoising
- Data compression
- Dimensionality reduction
- Anomaly detection
- Feature extraction

21. What is a Deep Generative Model?

Answer: A Deep Generative Model learns patterns in existing data and generates new data similar to the original data.

22. What is the difference between a generative model and a discriminative model?

Answer: A generative model can learn and generate new data, while a discriminative model learns to classify or predict labels.

23. What is a Variational Autoencoder (VAE)?

Answer: A VAE is a generative model that learns the probability distribution of data and can generate new similar samples.

24. What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?

Answer: A traditional Autoencoder mainly focuses on reconstruction, while a VAE learns a probabilistic latent space and can generate new data.

25. What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?

Answer: GANs consist of two neural networks:

- Generator: Creates fake data similar to real data.
- Discriminator: Identifies whether the data is real or fake.

The competition between them helps generate more realistic data.

PHOTO:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Subject Knowledge and Conceptual Understanding	30%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Technical Accuracy and Application of Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Problem-Solving and Analytical Thinking	20%	Innovative, practical, and feasible solution aligned with	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

		industry standards			
Communication Skills and Clarity of Explanation	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Confidence, Presentation and Professional Behaviour	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis